DP Biology – A1.1 (HL extension)

Where water came from

- Which features of water make life possible?
- What does living in water make easier and harder for organisms?

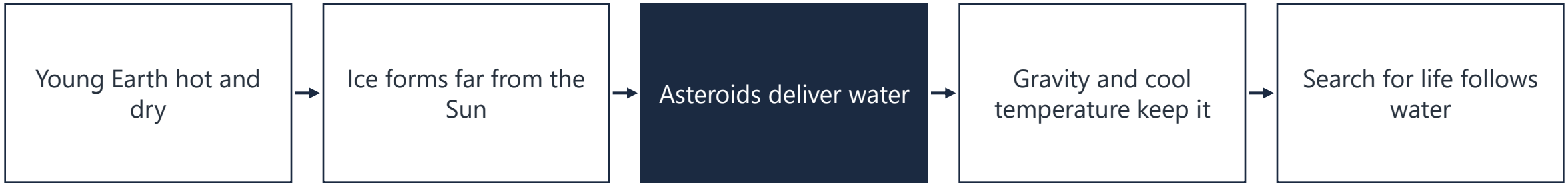
HL statements A1.1.7–8. Textbook pp. 11–13.
Image: the Earth from Apollo 17. NASA, public domain.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	A1.1.7	Evaluate the asteroid hypothesis for the origin of Earth's water, and explain why Earth kept it
2	A1.1.8	Explain the habitable zone and why the search for life beyond Earth starts with water

The big picture: water from space



Most of Earth's water probably came from **outside the planet**. Earth kept it because of its **gravity** and **temperature**.

Because life needs liquid water, the search for life elsewhere begins by looking for it.

Big question

Why does Earth have oceans while Venus and Mars do not?

A wet planet that formed dry

A1.1.7



Earth from Apollo 17. NASA, public domain.

The Earth formed about **4.5 billion years ago** close to the Sun, where it was too hot for water vapour to condense.

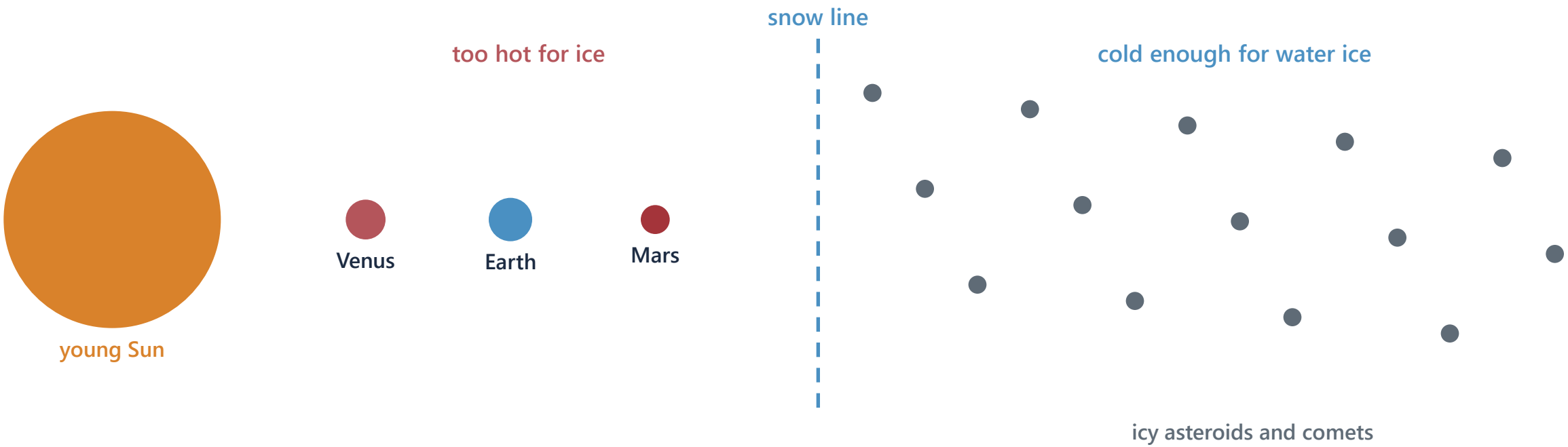
So most of its water probably arrived later from outside: it is **extraplanetary**.

Question

If the young Earth was too hot to hold water, how did oceans come to cover 70% of it?

The snow line

A1.1.7



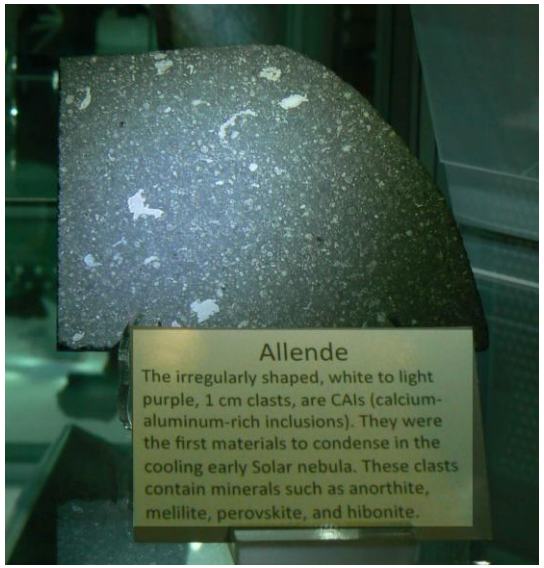
Key idea

Inside the snow line, water stayed as vapour and was blown away. Beyond it, water froze onto asteroids and comets that could later strike Earth.

The asteroid hypothesis



The asteroid Vesta. NASA/JPL, public domain.



Allende, a carbonaceous chondrite. Milo44, CC BY-SA 3.0.

Water was probably delivered by **asteroids** during the early bombardment of Earth.

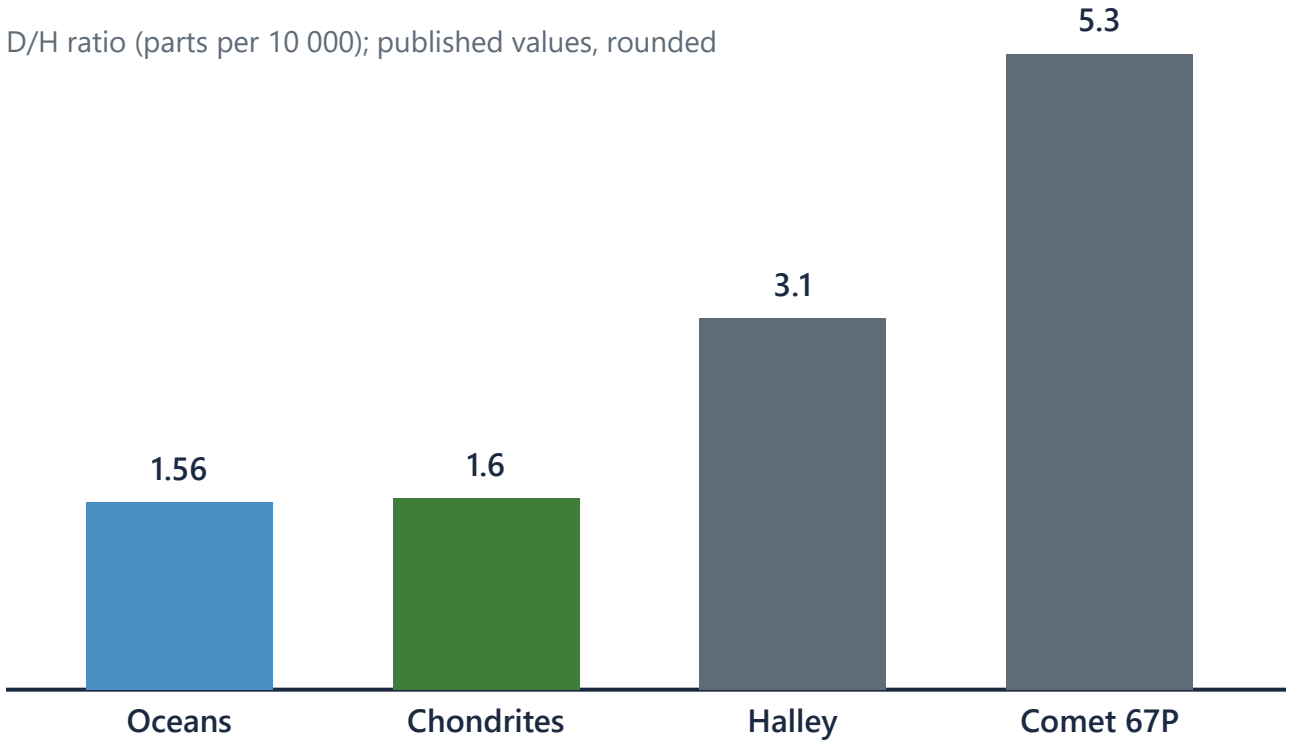
Fragments still fall as **carbonaceous chondrite** meteorites.

Evidence	What it shows
Water in meteorites	Some hold up to a fifth of their mass as water, in minerals
Isotope ratio	Their D/H ratio matches Earth's oceans

Deuterium: a fingerprint for water

A1.1.7

Deuterium is hydrogen with a neutron. The D/H ratio differs with where water formed, so it shows where water came from.



Conclusion

Chondrites match the oceans; most comets have far too much deuterium.

Working with data: weighing the evidence

A1.1.7

Source of water	D/H (per 10 000)
Earth's oceans	1.56
Carbonaceous chondrites	about 1.4 to 1.8
Most comets (e.g. Halley)	about 3
Comet 67P (Rosetta, 2014)	5.3
Comet Hartley 2	1.6

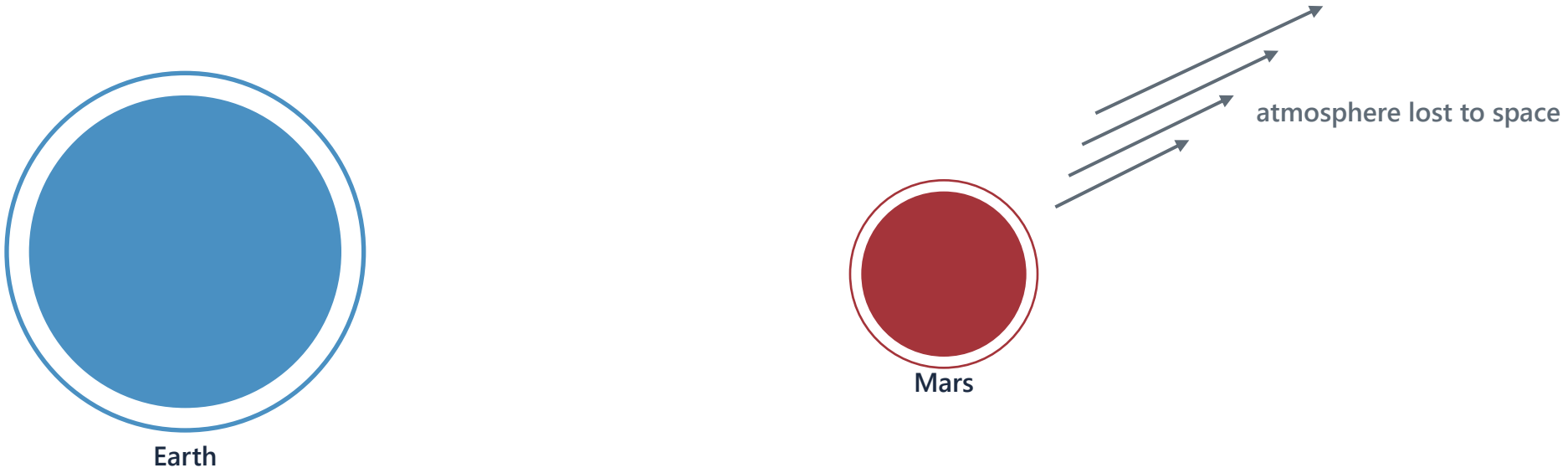
1. Which source matches ocean water best?
2. What does comet 67P suggest?
3. Why is one matching comet not enough?

Nature of science

Evidence is weighed, not counted: many comets disagree with one that fits.

Keeping water: gravity

A1.1.7



A planet needs enough **mass** for its **gravity** to hold an atmosphere and water vapour. Smaller Mars lost most of its atmosphere.

Plus temperature

Earth is cool enough for water vapour to condense into liquid oceans.

Why Earth kept its water

A1.1.7

	Venus	Earth	Mars
Distance from Sun	Closer	In the habitable zone	Further
Gravity	Similar to Earth	Holds an atmosphere	Weaker
Surface temperature	About 460 °C	Mostly 0–100 °C	Mostly below 0 °C
Water now	Lost to space	Liquid oceans	Frozen or underground

Big question

Earth has both enough **gravity** and a **temperature** low enough to condense water.

Practice: origin and retention

A1.1.7

Explain

Why the young Earth had little water of its own.

Evaluate

The asteroid hypothesis, using D/H data.

Predict

A planet like Earth but half its mass.

Use

snow line, extraplanetary, deuterium, gravity, condense

Check-in: part 1

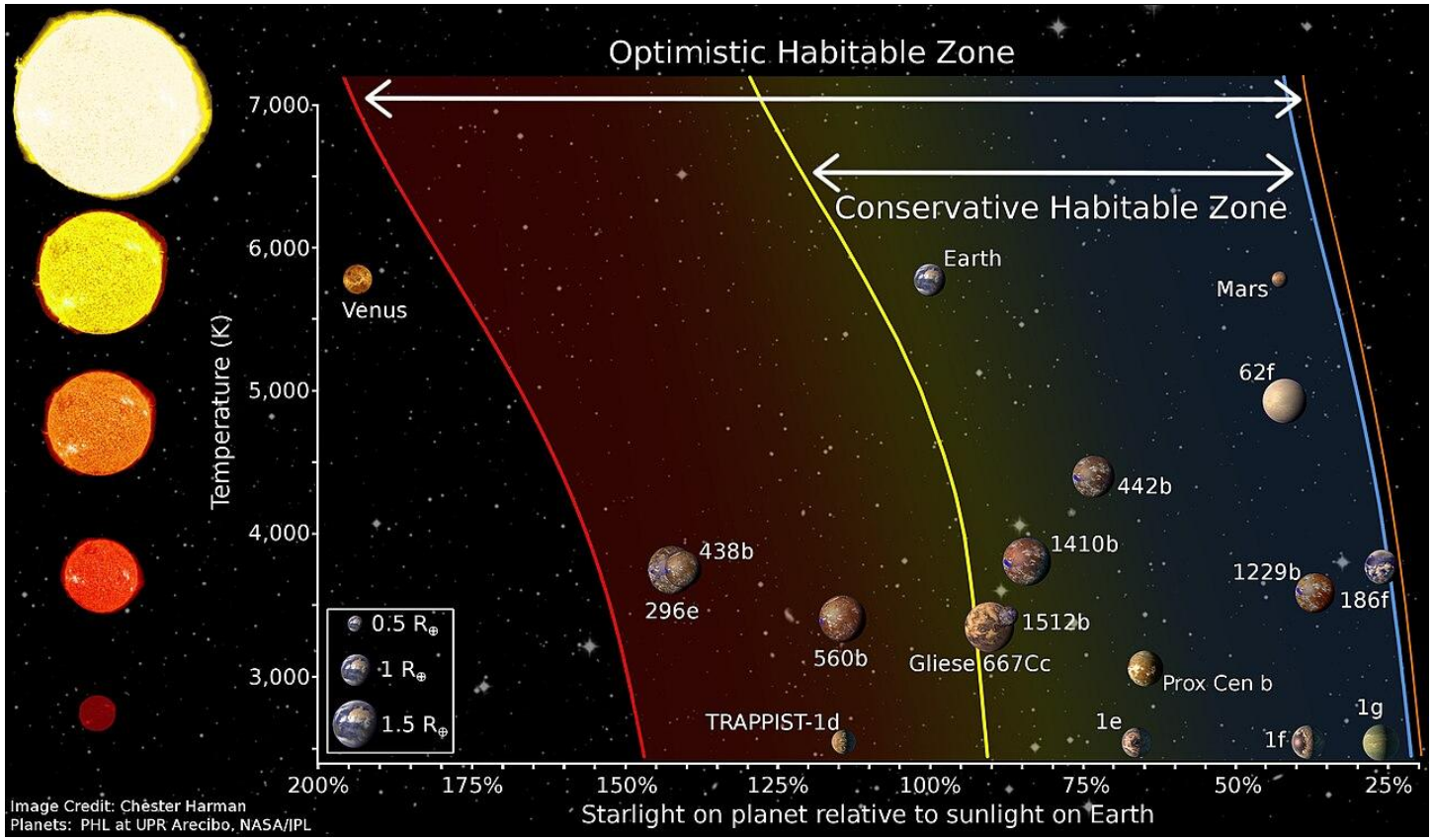
A1.1.7

1. Explain why most of Earth's water is extraplanetary.
2. State what the snow line is.
3. Outline two pieces of evidence for the asteroid hypothesis.
4. State two reasons Earth kept its water.
5. Explain why Mars lost most of its water.

How to use this

Answer without your notes, then check against slides 3–11.

The habitable (Goldilocks) zone



Habitable zones around stars of different temperatures. Chester Harman, CC BY-SA 4.0.

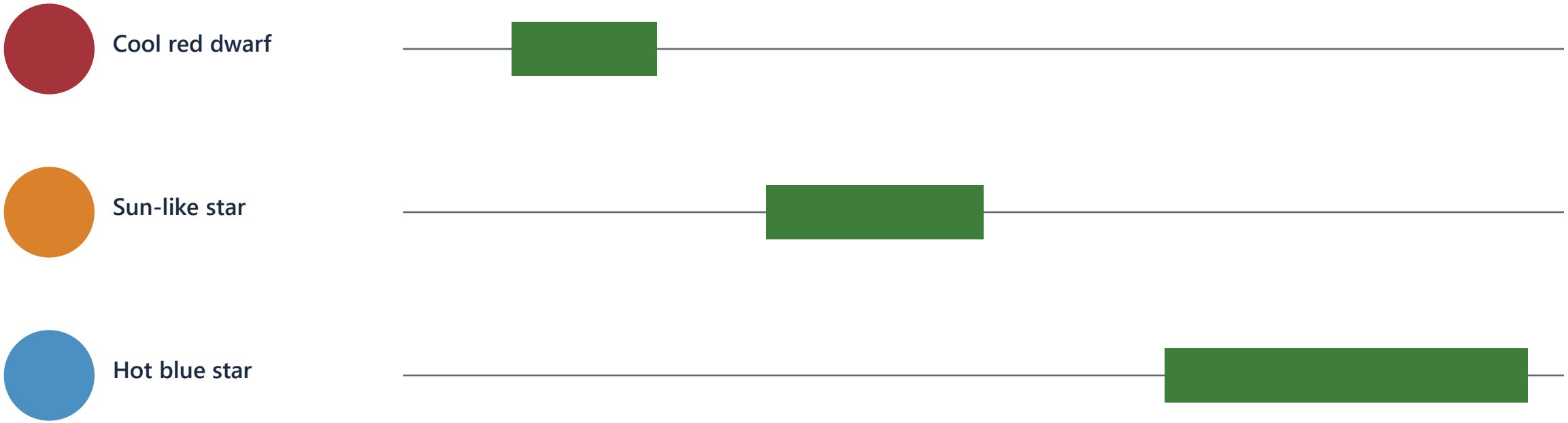
The **habitable zone** is the range of distances from a star where a planet's surface could hold **liquid water**.

Cooler stars have zones close in; hotter stars, further out.

Life as we know it needs liquid water, so finding water narrows the search for life.

Too hot, too cold, just right

A1.1.8



Key idea

Closer than the zone, water boils away; further out, it freezes.

Why the search for life starts with water

A1.1.8

Solvent	Medium	Transport
Dissolves the substances of metabolism	Reactions happen in water	Carries substances within organisms

All known life needs **liquid water**. Water is not proof of life, but no water makes life as we know it very unlikely.

Link
A1.1.1–6 (SL): hydrogen bonding, cohesion, solvent properties and thermal properties of water.

Finding water on distant planets

A1.1.8



Transit spectroscopy: starlight passing through a planet's atmosphere loses the wavelengths that water absorbs, leaving a signature telescopes detect.

Today

The James Webb Space Telescope measures exoplanet atmospheres this way.

Water in our own solar system

A1.1.8



Mars. Omarius257, public domain.

Body	Evidence of water
Mars	Old river valleys; ice at the poles and underground
Europa (Jupiter)	Likely ocean under its ice
Enceladus (Saturn)	Plumes of water, salts and simple organics

Think

Europa and Enceladus lie outside the Sun's habitable zone. How can they have liquid water?

Working with data: which planet could hold water?

A1.1.8

Invented planets for practice; realistic values.

Planet	Zone of its star	Mass (Earth = 1)	Atmosphere?
P	Inside	0.1	No
Q	Inside	1.2	Yes
R	Too close	1.0	Yes
S	Too far	5	Yes

1. Which planet is the best target? Why?
2. Why is P a poor target despite its position?
3. Suggest what may have happened to R's water.

In the zone is not enough

A1.1.8

Size

Enough gravity to hold an atmosphere

Protection

A magnetic field against the stellar wind

Time

Billions of years for life to arise

Discuss

A planet in the habitable zone might still have no life. Use these three cards to explain why.

Check-in: part 2

A1.1.8

1. Define the habitable zone.
2. Explain why cooler stars have closer habitable zones.
3. Explain why the search for life starts with water.
4. Outline transit spectroscopy.
5. State why being in the zone does not guarantee life.

How to use this

Answer without your notes, then check against slides 13–19.

Common mistakes to avoid

Not this	But this
Earth’s water mostly came from comets.	D/H evidence points mainly to asteroids.
Any planet in the zone has life.	The zone means liquid water is possible, no more.
Mars never had water.	It once had liquid water; now frozen or underground.
Only temperature keeps Earth’s water.	Gravity and temperature both matter.

Summary: water beyond Earth

A1.1.7–8

	Key idea
A1.1.7	Young Earth dry; water delivered by asteroids
A1.1.7	Evidence: chondrite water with matching D/H
A1.1.7	Kept by gravity and condensing temperatures
A1.1.8	Habitable zone: liquid water possible
A1.1.8	Search for life starts with water

Check that you can...

1. **Explain** why the young Earth had little water
2. **Evaluate** the asteroid hypothesis using D/H data
3. **Explain** how gravity and temperature kept Earth's water
4. **Define** the habitable zone
5. **Outline** transit spectroscopy
6. **Explain** why the search for life starts with water

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Give one piece of evidence that asteroids brought water to Earth.
2. State two reasons Earth has kept its water.
3. Explain why the search for life beyond Earth focuses on water.

Image credits

The Earth seen from Apollo 17.jpg – NASA/Apollo 17 crew; taken by either Harrison Schmitt or Ron – Public domain

Vesta in natural color (cropped).jpg – NASA / JPL / MPS / DLR / IDA / Björn Jónsson – Public domain

Allende meteorite slice ASU.jpg – Milo44 – CC BY-SA 3.0

Diagram of different habitable zone regions by Chester Harman.jpg – Chester Harman – CC BY-SA 4.0

OSIRIS Mars in true color.png – Omarius257 – Public domain

All from Wikimedia Commons (commons.wikimedia.org, search the file name). Some images are cropped. CC licences: creativecommons.org/licenses